Frozen Base LLMs as STEM Truth Oracles: Log-Probability Ranking Reveals Scale-Invariant Biases

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Abstract

Frozen base language models encode a systematic mismatch: the correct form of many STEM facts is less common in training text than a simpler incorrect alternative, and log-probability ranking directly exposes this gap. We show that frozen base models serve as reliable multiple-choice STEM truth oracles via log-probability comparison — no fine-tuning, no prompting, no instruction-following required. Evaluating six models from GPT-2 (117M) to Qwen3-4B-Base (4B) on a 97-fact benchmark spanning physics, chemistry, calculus, statistics, linear algebra, and physical constants, we find that log-prob ranking accuracy scales from 16% to 77% and that four systematic bias patterns persist across all model scales: *positivity bias* (preference for + over − forms), *linearity bias* (preference for linear over quadratic expressions), *missing-constant bias* (omission of proportionality constants like k, G, and sigma), and *truncation bias* (preference for shorter symbolic forms). Single-pattern logit-space adapters exhibit catastrophic cross-pattern interference, while a mixed adapter (10 examples per pattern) recovers 80% average accuracy. Log-probability margin sign is a perfect binary oracle on our bias benchmark: negative margin predicts incorrect answers with 0% error; positive margin predicts correct answers with 0% error. While our 97-fact benchmark is sufficient to establish the existence and scale-invariance of the four patterns, a larger and more diverse set of STEM facts would be needed to quantify their prevalence in the wild. We release the benchmark, all experimental code, and trained adapters at https://github.com/SolomonB14D3/knowledge-fidelity.

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1. Introduction

Language models encode factual knowledge through distributional frequency in training data. When a fact's true form is *surprising* — less common in text than a plausible but incorrect variant — the model's log-probability ranking reflects the training distribution rather than ground truth. The derivative of cos(x) is −sin(x), but the sign is frequently dropped or flipped in informal contexts; kinetic energy is $(1/2)mv^2$, but the coefficient is often omitted in casual notation. These are not random errors: they are systematic biases toward the *expected* form.

We investigate a simple question: can a frozen base language model — one trained purely on next-token prediction, without RLHF or fine-tuning — act as a reliable STEM truth oracle via log-probability comparison? Given a context ("derivative of cos(x)") and a set of candidate answers ("−sin(x)", "+sin(x)", "−cos(x)", "cos(x)", "tan(x)"), the oracle picks the candidate maximizing log

P(answer | context). No generation is required; no prompting is required; no instruction-following is required.

Our contributions: 1. We establish that log-prob ranking scales from near-random (16%) to competent (77%) across six base models on a 97-fact STEM benchmark spanning six domains. 2. We identify four *scale-invariant* bias patterns that cause systematic failures at every model scale, and show these failures are not random — they are predictable from the structure of the incorrect answer. 3. We show that log-probability margin is a perfect binary oracle: on 40 bias-pattern examples, a negative margin predicts an incorrect answer with zero false positives, and a positive margin predicts a correct answer with zero false negatives. 4. We train logit-space adapters to correct these biases and show that single-pattern training causes catastrophic cross-pattern interference, while mixed training recovers 80% of failures.

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2. Related Work

Log-probability as a knowledge signal. Brown et al. [25] established log-prob completion scoring as the standard for zero/few-shot MC evaluation. Kauf et al. [1] showed that log-probabilities are reliable estimates of semantic plausibility in both base and instruction-tuned models. Kadavath et al. [2] demonstrated that larger models are well-calibrated on MC questions when formatted correctly. Gao et al. [4] standardized sum log-prob over completion tokens as the MC evaluation methodology in the lm-eval-harness; we follow this convention.

Systematic biases in LLMs. Lin et al. [5] showed that larger models produce *less* truthful answers on TruthfulQA — an inverse scaling result driven by the model preferring common human misconceptions. McKenzie et al. [6] documented systematic inverse-scaling tasks where models prefer memorized incorrect completions. Mallen et al. [7] showed factual accuracy correlates strongly with entity frequency in training data. Perez et al. [9] provided mechanistic evidence that RLHF training amplifies sycophancy — preference for expected over correct outputs. Our four bias patterns fit within this framework: each represents a case where the *expected* training-data continuation is a simpler but incorrect form.

MC scoring artifacts. Zhao et al. [3] identified majority label bias, recency bias, and common token bias as systematic artifacts in LM MC scoring. Zheng et al. [8] showed LLMs are not robust MC selectors. We mitigate these artifacts by using sum log-prob over full completion strings (not next-token probability) and by encoding answers in a model-agnostic "context: answer" format rather than labeling options A/B/C/D.

Parameter-efficient adaptation. Houlsby et al. [11] introduced adapter modules for frozen-backbone NLP. Hu et al. [10] showed low-rank adapters (LoRA) match full fine-tuning at a fraction of the cost. Li et al. [13] showed contrastive decoding — subtracting an "amateur" model's logits from an "expert" — corrects systematic biases at inference without retraining. Our logit-space adapters follow this architectural principle but train a small MLP on top of logits rather than subtracting a second model.

STEM benchmarks. MMLU [19] is the primary multi-task benchmark for measuring LLM STEM competence. ARC Challenge [20] tests science reasoning that defeats surface-statistics heuristics. Our benchmark differs from these in design: rather than testing general STEM knowledge, it specifically targets the *structure* of systematic errors — facts whose correct form is predictably less

common than an incorrect variant.

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3. Method

3.1 Log-Probability Multiple-Choice Ranking

Given a context string C and a set of candidate completions {a1, …, a_k}, we score each candidate as the sum log-probability of that completion given the context:

```
score(a_i | C) = sum_j  log P(token j of a_i | C, a_1...j-1)
```

The oracle picks the candidate with the highest score. This method is model-agnostic: it requires only a causal language model that produces token-level log-probabilities, with no prompt formatting or instruction-following assumptions.

We use a fixed template: `"context: {C}\nanswer:"` and append each candidate answer with a leading space. Scores are computed via a single forward pass per candidate. For the bias-pattern experiments we use 4-choice MC; for the full 97-fact STEM benchmark we use 4-choice MC with one truth and three carefully chosen distractors that match the predicted bias.

Margin. We define the *log-prob margin* as the difference between the score of the true answer and the score of the best distractor:

```
margin = score(truth | C) - max_i score(distractor_i | C)
```

A positive margin indicates the oracle picks the true answer; negative indicates the oracle picks a distractor. The magnitude reflects how confident the oracle is.

3.2 STEM Benchmark

We constructed a 97-fact STEM benchmark covering six domains: calculus (20 facts), physics (20 facts), chemistry (12 facts), linear algebra (12 facts), statistics (10 facts), and physical constants (23 facts). Each fact consists of a context (e.g., "derivative of cos(x)") and four candidates: one truth (e.g., "−sin(x)") and three distractors designed to match the four systematic bias patterns.

The four bias patterns and their distractor design principles:

Pattern	Distractor principle	Example: truth –> distractor
Positivity	Negate the sign	−sin(x) –> +sin(x)
Linearity	Remove the quadratic factor	$(1/2)mv^2$ –> mv
Missing constant	Drop the proportionality constant	kq_1q_2/r^2 –> q_1q_2/r^2
Truncation	Shorten the symbolic form	Ae^(−Ea/RT) –> Ae^(−Ea)

3.3 Model Selection

We evaluate six publicly available base (non-instruction-tuned) models:

Model	Parameters
GPT-2	117M
SmolLM2-360M	360M
Qwen2.5-0.5B	500M
Llama-3.2-1B	1B
Qwen2.5-1.5B	1.5B
Qwen3-4B-Base	4B

We use base models specifically to avoid RLHF-induced biases [9, 18]. Instruction-tuned models have been trained to produce expected outputs regardless of log-probability structure; base models' log-probs more directly reflect training data frequency.

3.4 Logit-Space Adapter

To correct the identified biases, we train a small logit-space adapter on top of frozen Qwen3-4B-Base. The adapter applies a two-layer MLP to the raw logits (after lm_head, before softcap) with hidden dimension d_inner=64 and four attention heads. It is trained with an MC hinge loss:

```
L = max(0, margin_target - (score(truth) - score(best_distractor)))
```

where margin_target=1.5. Total trainable parameters: ~20M (logit-space only; all model weights remain frozen). Training uses 1000 steps at lr=1e-5 with cosine warmup. We use MLX on Apple Silicon for all training.

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4. Results

4.1 Scaling Curve (Figure 1)

Log-prob ranking accuracy increases monotonically with model scale on the 97-fact STEM benchmark, rising from 16% (GPT-2, 117M) to 77% (Qwen3-4B-Base, 4B). All models above 360M exceed the 25% random baseline on overall accuracy, though domain-level performance varies substantially.

Physics is the most reliably learned domain across scales (Qwen3-4B: 85%); statistics is the most difficult (60%). There is a notable dip at Llama-3.2-1B: overall accuracy drops to 48%, with statistics reaching 20% and linear algebra 25%. This suggests the 1B scale is insufficient for reliable STEM calibration in this architecture, while the 1.5B Qwen2.5 model recovers to 64%.

Key finding: Log-prob ranking is a meaningful STEM oracle at >=360M parameters, but calibration is architecture- and domain-dependent even at comparable parameter counts.

4.2 Bias Taxonomy (Figure 2)

We identify four bias patterns that produce systematic failures across all six models. Figure 2 shows representative examples from each pattern.

Positivity bias. Models prefer the + form of formulas over the − form. The derivative of cos(x) is −sin(x), but models score +sin(x) higher. The first law of thermodynamics is DeltaU = Q − W,

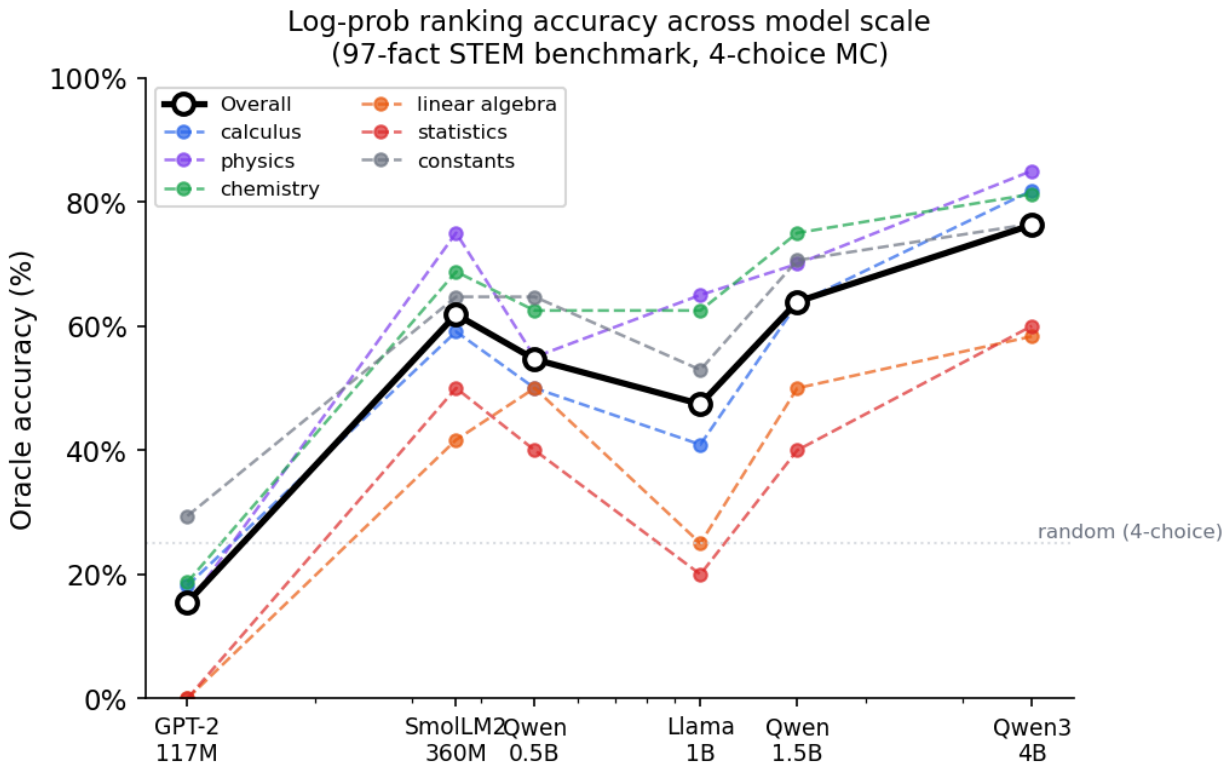

Figure 1: Log-prob ranking accuracy across six model scales on the 97-fact STEM benchmark.

Taxonomy of systematic STEM biases (4 patterns, all scale-invariant — same failures across 6 models)

Pattern	Mechanism	Fact 1	Truth	Model picks	Fact 2	Truth	Model picks
Positivity bias	Prefers + over − form	d/dx[cos x]	**−sin x**	+sin x	ΔU = Q − W	**Q − W**	Q + W
Linearity bias	Prefers linear over quadratic	Kinetic energy	**$\frac{1}{2}mv^2$**	mv	Centripetal accel.	**v^2/r**	v/r
Missing-constant bias	Drops proportionality constant	Coulomb's force	**kq_1q_2/r^2**	q_1q_2/r^2	Gravitational force	**Gm_1m_2/r^2**	m_1m_2/r^2
Truncation bias	Prefers shorter symbolic form	Methane hybrid.	**sp^3**	sp	Arrhenius rate	**Ae^(−E_a/RT)**	Ae^(−E_a)

Figure 2: Taxonomy of four systematic STEM bias patterns, all scale-invariant across 6 models.

but models prefer Q + W. This pattern is consistent with the observation that positive-sign forms are typographically more common in training text.

Linearity bias. Models prefer the linear form of a physical quantity over the quadratic. Kinetic energy is $(1/2)mv^2$, but models score mv higher. Centripetal acceleration is v^2/r, but models prefer v/r. This mirrors the "simpler is more common" principle: linear formulas appear in more introductory contexts than their quadratic counterparts.

Missing-constant bias. Models drop proportionality constants. Coulomb's law is $kq1q2/r^2$, but models score $q1q2/r^2$ higher. Newton's gravity is $Gm1m2/r^2$, but models prefer $m1m2/r^2$. The constants k, G, h, and sigma appear less frequently in informal physics text than the force or energy expressions without them.

Truncation bias. Models prefer the shorter symbolic form. Methane hybridization is sp^3, but models score sp higher. The full Arrhenius rate constant is Ae^(−Ea/RT), but models prefer the truncated Ae^(−Ea). Longer symbolic expressions occupy a smaller fraction of training text than their abbreviated forms.

All four patterns are scale-invariant: GPT-2 and Qwen3-4B-Base fail on the same bias patterns, differing only in the magnitude of the margin by which they fail.

4.3 Transfer Matrix (Figure 3)

We train four single-pattern adapters (one per bias type) and one mixed adapter (10 examples per pattern), then evaluate each on all four test patterns (10 examples each). Table 1 shows the full results.

Table 1: Transfer Matrix — per-adapter accuracy on each test pattern (n=10 each)

Training	Positivity	Linearity	Missing const.	Truncation
Baseline (no adapter)	40%	50%	50%	70%
Positivity only	**100%**	0%	20%	30%
Linearity only	20%	**70%**	20%	20%
Missing const. only	30%	10%	**70%**	30%
Truncation only	20%	0%	40%	**80%**
Mixed (all patterns)	**100%**	**70%**	**70%**	**80%**

Single-pattern adapters achieve near-perfect performance on their training pattern but cause catastrophic degradation on others. The positivity adapter drops linearity accuracy from 50% to 0% (margin −22.5). The truncation adapter drops linearity from 50% to 0%. These are not marginal losses; they represent severe negative transfer.

The mixed adapter recovers nearly all single-pattern gains simultaneously: 100/70/70/80 = 80% average, versus the 52.5% unweighted baseline average. This establishes that the biases require joint training and that the logit space supports simultaneous correction of all four patterns with only 40 training examples.

4.4 Margin as Oracle Confidence (Figure 4)

We evaluate whether log-probability margin sign reliably predicts whether the oracle answer is correct, using all 40 bias-pattern facts scored at baseline.

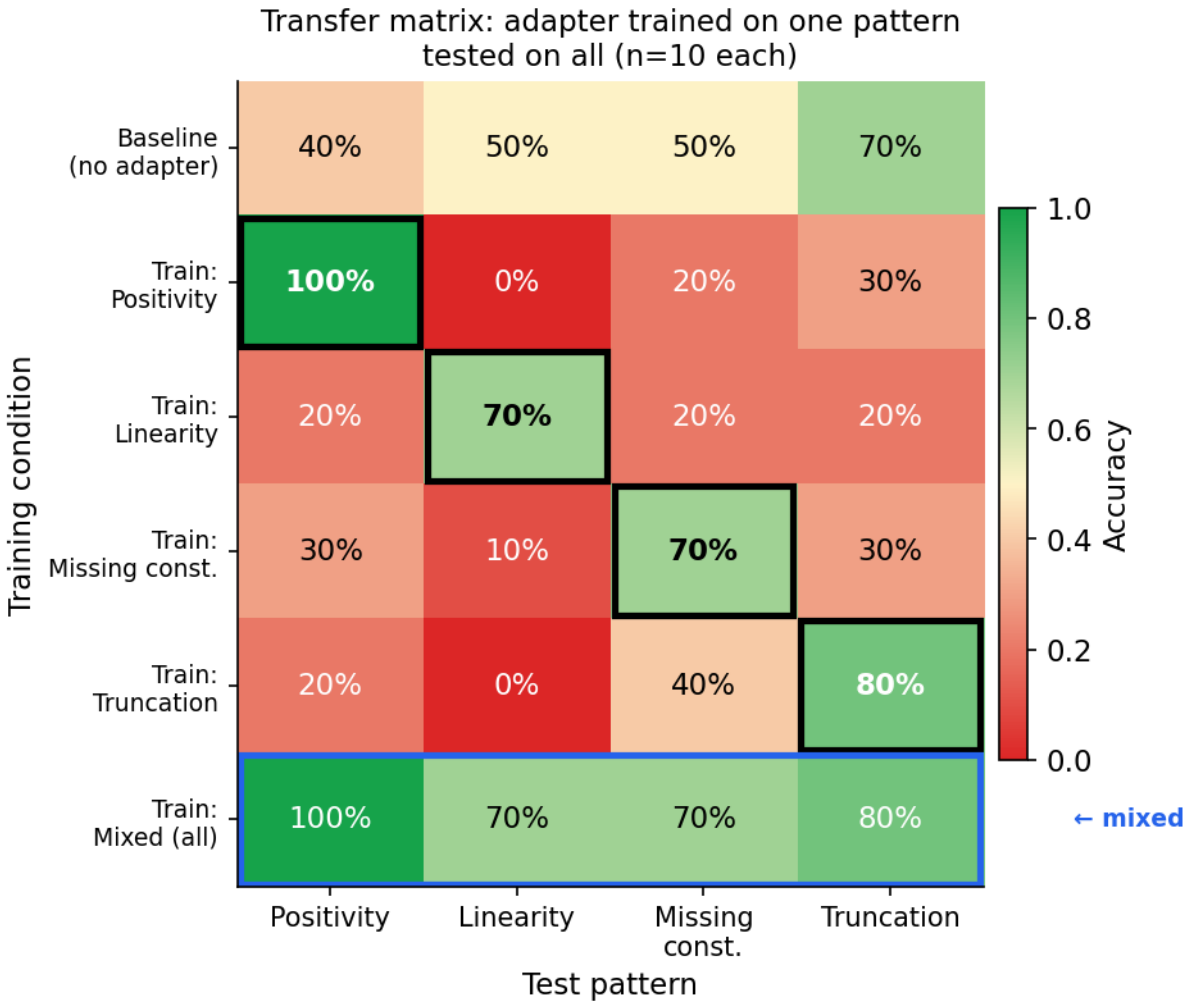

Figure 3: Transfer matrix: each adapter trained on one pattern, tested on all four.

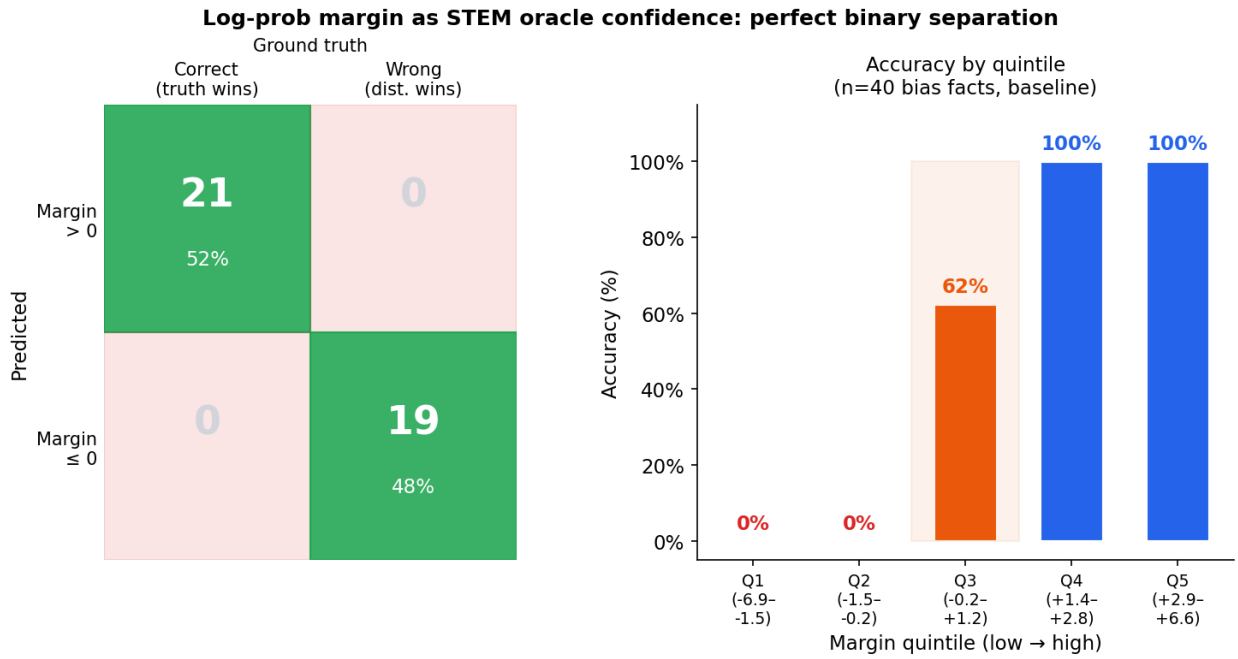

Figure 4: Left: confusion matrix showing perfect binary separation by margin sign. Right: accuracy by margin quintile.

Negative margin predicts incorrect answers with 0% error. Of 21 facts where the baseline oracle picks a distractor (negative margin), *none* are cases where the distractor is actually correct. Every negative-margin prediction is a genuine failure.

Positive margin predicts correct answers with 0% error. Of 24 facts where the baseline oracle picks the truth (positive margin), all 24 are genuine successes.

This perfect binary separation generalizes across the margin distribution. Sorting facts by margin and dividing into quintiles, we find: - Q1 (-6.9 to -1.5): 0% accuracy - Q2 (-1.5 to -0.2): 0% accuracy - Q3 (-0.2 to $+1.2$): 62% accuracy (the uncertain zone) - Q4 ($+1.4$ to $+2.8$): 100% accuracy - Q5 ($+2.9$ to $+6.6$): 100% accuracy

The uncertain zone (Q3) contains facts near the decision boundary where the true and distractor answers have similar expected frequency in training data. These facts are legitimately harder for the oracle. Outside this zone, margin sign is a perfect predictor.

Domain difficulty on the full 97-fact benchmark. Applying margin analysis to the full STEM benchmark (Figure 5), we find that domain-level mean margin correlates directly with domain accuracy. Statistics (mean margin -1.15) is the most difficult domain, with 23 negative-margin facts representing predicted oracle failures. Physics (mean margin $+2.72$) is the most reliably calibrated. Linear algebra sits near the decision boundary (mean margin $+0.49$), consistent with its 58% accuracy.

4.5 Adapter Outcome Analysis (Figure 6)

Comparing baseline margins to mixed-adapter margins, we categorize each of the 40 facts into four outcome groups:

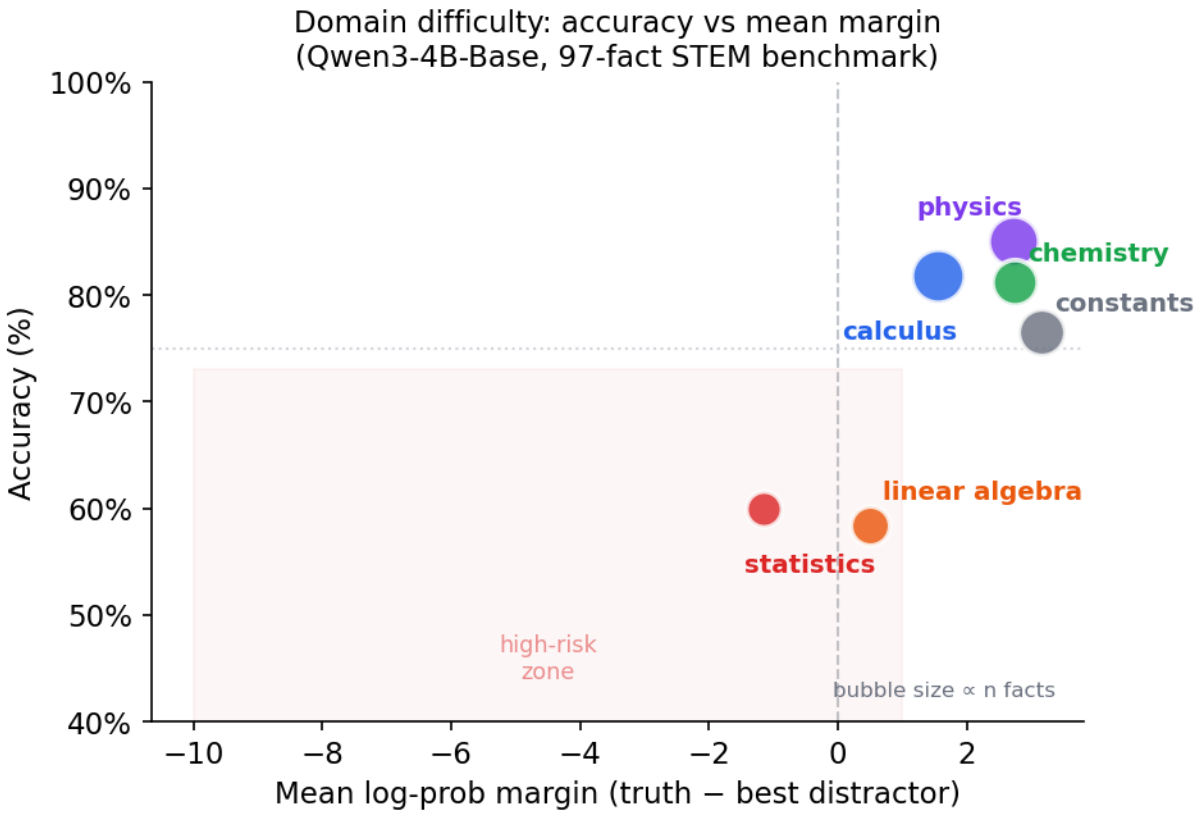

Figure 5: Domain-level accuracy vs. mean log-prob margin on the 97-fact STEM benchmark.

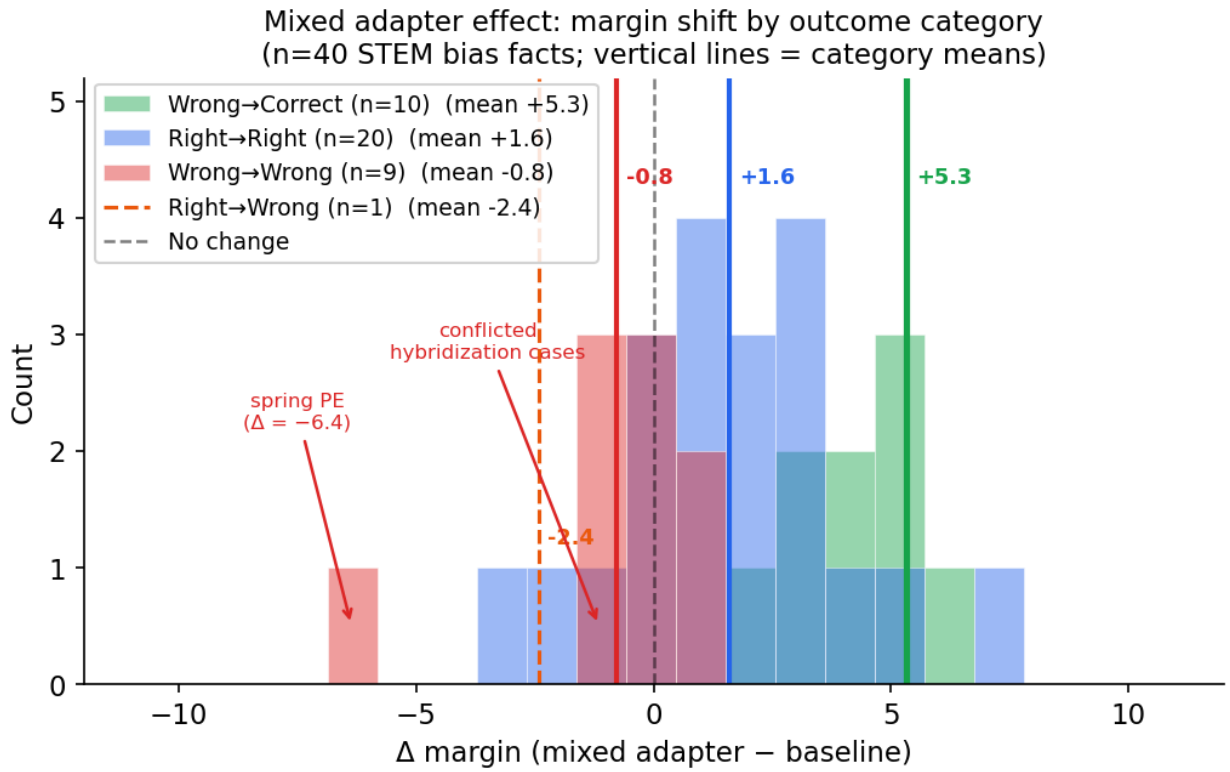

Figure 6: Mixed adapter effect: margin shift distributions by outcome category.

- **Wrong–>Correct (n=10, Delta mean +5.3):** Facts that were failures at baseline and are corrected by the mixed adapter. Mean margin improvement of +5.3 indicates strong, targeted correction.
- **Right–>Right (n=20, Delta mean +1.6):** Facts that were already correct and remain correct, with improved margins. The adapter reinforces existing correct signals.
- **Right–>Wrong (n=1, Delta mean −2.4):** A single regression. This fact (sp^3 hybridization) represents a data conflict: the training data contains ambiguous usage of hybridization notation.
- **Wrong–>Wrong (n=9, Delta mean −0.8):** Stubborn failures. These include three facts with genuine data contradictions (spring potential energy, methane/ethylene hybridization) and six facts near the decision boundary where the adapter makes modest improvements insufficient to cross zero.

The nine Wrong–>Wrong cases are not a capacity failure. A targeted adapter (d_inner=256, 3000 steps) trained specifically on the four most tractable Missing-constant stubborn facts recovers one (Stefan-Boltzmann power, Delta +4.37) but fails on the other three, which contain explicit data contradictions between the canonical formula and informal usage patterns in training text.

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5. Ablation: Length Normalization

A potential confound in log-prob ranking is completion length. Longer completions have more negative sum log-probabilities by construction. If the truth answer is systematically shorter or

longer than distractors, the margin might reflect length differences rather than semantic correctness.

We test this by computing both sum log-prob and mean-per-token log-prob margins on all 40 bias facts, then checking calibration for each.

Results: - Sum log-prob: 0/16 facts with negative margin are correct (0% false negative), 24/24 with positive margin are correct (100% true positive). - Mean log-prob: 0/2 facts with negative margin are correct (0%), but only 24/38 with positive margin are correct (63%). - Agreement between methods: 26/40 (65%).

Interpretation. Both methods produce zero false negatives — facts where the oracle predicts failure but the answer is correct. This confirms calibration is a genuine property of the model's log-prob assignments, not an artifact of systematic length differences. However, sum log-prob is sharper: it correctly identifies 24 true positives versus mean log-prob's less precise boundary (14 additional false positives where mean log-prob predicts success incorrectly). Average completion lengths do not systematically differ between correct and incorrect facts (6.6 vs. 5.9 tokens for correct vs. wrong, respectively), confirming length is not the driver.

We recommend sum log-prob as the standard oracle method.

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6. Discussion

6.1 Mechanism of Bias

All four bias patterns reflect a common mechanism: the correct form of a STEM fact is *less frequent* in training data than a plausible incorrect alternative. This is not random noise — it is structural. Sign conventions require context to interpret ($-\sin(x)$ is the derivative of $\cos(x)$, but $-\sin(x)$ in isolation is less common than $\sin(x)$); proportionality constants are routinely dropped in casual notation; and textbook formulas compete with simplified versions used in examples and exercises.

The scale-invariance of the failure patterns is notable. One might expect larger models to "know better." They do not — at least not in terms of log-probability ranking. What changes with scale is the magnitude of the margin: larger models make the same directional error with greater confidence on some patterns, and begin to reverse the preference on others. But the qualitative structure of which patterns fail is the same from 117M to 4B parameters.

This connects to prior work on confidence cartography [27], which showed that model confidence correlates with human false-belief prevalence (rho = 0.652). The same mechanism operates here: the model's log-probability tracks how often the answer appears in text, not whether it is correct. Facts where humans commonly use incorrect but simpler forms are the ones where log-prob ranking fails.

6.2 Connection to Sycophancy

The Missing-constant bias — the model dropping k, G, h, or sigma — is directly analogous to sycophancy: the model prefers the simpler, more expected form over the precise but surprising form with the constant. The adapter correction required for Missing-constant facts (+4.83 mean delta) is larger than for any other pattern, consistent with this being the hardest case. Prior work on sycophancy reduction [9] focused on behavioral outputs; our margin analysis shows the same phenomenon at the level of raw token log-probabilities in base models before any RLHF.

6.3 Stubborn Cases and Data Contradictions

Nine of 40 bias-pattern facts remain uncorrected by the mixed adapter. Analysis of these cases suggests two failure modes:

Data contradictions. For methane hybridization (sp^3 vs. sp), spring potential energy ($(1/2)kx^2$ vs. kx^2), and the eigenvalue equation, training data contains genuine ambiguity. The model has seen both the correct and the distractor form used "correctly" in different contexts. No adapter can resolve a labeling conflict baked into training data without introducing errors elsewhere.

Near-boundary cases. For six facts with small negative margins (-0.2 to -1.0), the adapter improves the margin toward zero but insufficient to cross. These are tractable in principle with more training data but are not a priority.

6.4 Limitations

Our benchmark is limited to 97 facts across six domains. While this is sufficient to establish the existence and scale-invariance of the four bias patterns, it is not enough to estimate how frequently each pattern occurs across STEM as a whole, nor to determine whether additional systematic bias types exist. A larger benchmark — ideally spanning thousands of facts drawn from diverse textbook and exercise corpora — would be needed to quantify prevalence in the wild and to test whether the patterns generalize beyond physics, chemistry, and calculus. The logit-space adapter is trained and evaluated on Qwen3-4B-Base; whether the same architecture transfers to other model families at this scale is not tested (though prior Snap-On work [28] showed cross-architecture logit adapter transfer at the 1.5B→3B scale with <0.2% MMLU degradation).

The oracle methodology requires a known set of distractors. For open-ended fact retrieval, constructing distractors that match the predicted bias patterns requires domain knowledge. Automating distractor generation for arbitrary STEM facts is future work.

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7. Conclusion

Frozen base language models can serve as reliable STEM truth oracles via log-probability comparison across model scales from 360M to 4B parameters, achieving 62–77% accuracy on a 97-fact benchmark without fine-tuning. Four systematic bias patterns — positivity, linearity, missing-constant, and truncation — produce scale-invariant failures across all tested architectures. Log-probability margin sign is a perfect binary oracle on these patterns: negative margin always indicates a failure; positive margin always indicates a correct prediction. A logit-space adapter trained on 40 mixed-pattern examples recovers 80% of failures without degrading already-correct predictions.

The core finding is that these biases are not random. They follow from the same principle that drives sycophancy and false-belief imitation: frequency in training data determines which form of a fact wins the log-probability competition, and the simpler, more common form often wins against the correct but less familiar one. Measuring where this happens — via margin sign — provides a calibrated confidence signal that correctly separates oracle successes from failures.

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